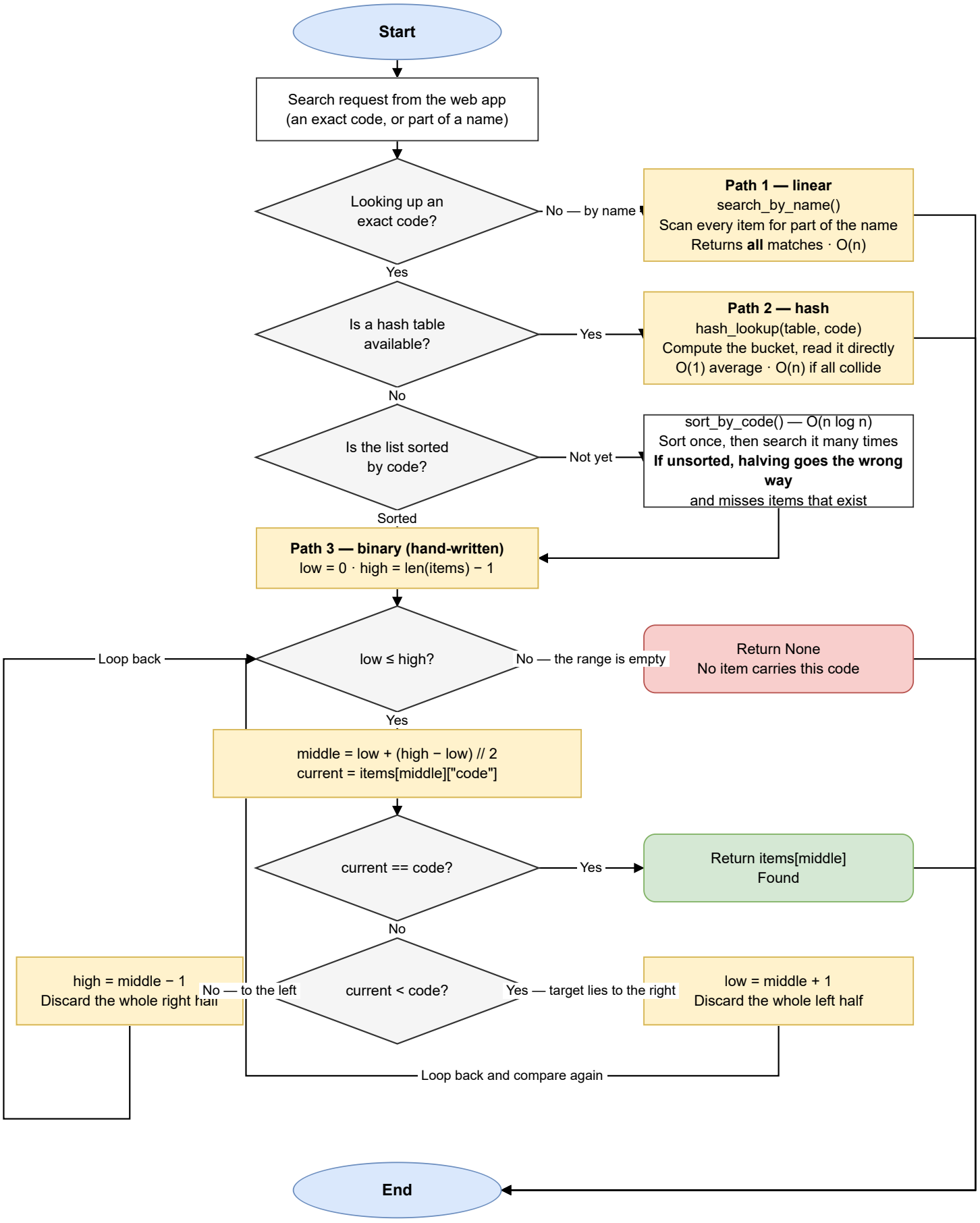
Flowchart 4 — Search: three paths (hash · binary · linear)

search.py · The path is chosen by the kind of question asked; binary_search, the hand-written one, is drawn in full



Why one round halves the work Each comparison halves what is left to examine, so 1,000 items need about 10 comparisons ($2^{10} = 1{,}024$), against roughly 500 on average for a linear scan · Written as a loop rather than recursion, so it adds no stack frames: memory is O(1), not O(log n)